

Uncertainty in Machine Learning - WBAI054-05

Assignment 1

Total points: **40**
Starting date: 21 April 2026
Submission deadline: **23:59, 6 May 2026**

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General guidelines:

- The tasks are targeted at groups of 2-3 students. Please make sure that the load is well divided: every student should contribute. In the case of not equal contributions that grade will be divided accordingly.
- Please take advantage of the tutorial sessions to ask your questions about the tasks.
- Provide a (short but comprehensive) explanation of what you are doing for each task.
- A reviewer should be able to understand plots independently; be sure to label axes, a legend for colors, use an easily readable font size, etc.
- Refer to all plots, tables, code blocks, etc. in your report.
- Submit the report as a single PDF file through Brightspace.
- Submit all code in a zipped file named *Code_Group_GroupNumber*, e.g., *Code_Group_20* through Brightspace.

Part I - Uncertainty in Medical AI

Maximum obtainable points: **20**

We have mentioned how uncertainty is very important in medical applications. For this part you will find a paper in the research medical AI literature that uses uncertainty quantification, **choose one** of the following constraints:

- The paper uses active learning with some form of uncertainty quantification.
- The paper mainly deals with aleatoric/data uncertainty.

Provide a description about the following from your paper:

1. The machine learning task (classification, regression, segmentation, etc).
2. Which uncertainty estimation method is used, or how uncertainty is used or modelled.
3. What kind of uncertainty is modelled/output (aleatoric, epistemic, or both).

To search research papers, you can use Google Scholar, MedPub, or look at proceedings for the following conferences: MIDL or MICCAI. For each point above, provide a short description.

Part II - Validating the Central Limit Theorem

Maximum obtainable points: **20**

In the first set of lectures we covered multiple properties of probability distributions, in the slides titled "Probability Theory and Probability Distributions". In this part you will answer the following two questions related to probability distributions.

Q1. Explain in your own words and/or give intuitions on why the Central Limit Theorem justifies using Gaussian distributions frequently in uncertainty quantification via machine learning models.

Q2. When might the assumptions made by the Central Limit Theorem be broken? Given at least one concrete example.

The answers for these questions should be conceptual, you can draw knowledge from the first two lectures, and if you use any arguments from the literature, please explain them in detail and give a proper citation.